

Quadratic Equations – Class XI

Complete Extraction: All Questions + Final Answers Only

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1 5.1 Solving Quadratic Equations by Formula

1.1 Illustrative Examples

Example 1. Solve the following quadratic equations:

(i) $3x^2 - 4x - 4 = 0$

(ii) $4x^2 - 2x + \frac{1}{4} = 0$

(iii) $3x^2 - 4x + \frac{20}{3} = 0$

Example 2. Solve the following equations:

(i) $x^2 + 3x + 9 = 0$

(ii) $27x^2 - 10x + 1 = 0$

Example 3. Solve the following equations:

(i) $\sqrt{3}x^2 - \sqrt{2}x + 3\sqrt{3} = 0$

(ii) $x^2 + \frac{x}{\sqrt{2}} + 1 = 0$

Example 4. Find the roots of the equation $x^2 + x - (a + 2)(a + 1) = 0$.

Example 5. Solve the equation $2x^2 - 5x + 2 = 0$ by the method of completing the squares.

Example 6. Find two numbers such that their sum is 6 and the product is 14.

Example 7. Three consecutive natural numbers are such that the square of the middle number exceeds the difference of the squares of the other two by 60. Assuming the middle number to be x , form a quadratic equation and find the three numbers.

Example 8. A two digit number is four times the sum and twice the product of its digits. Find the number.

Example 9. A factory kept increasing its output by the same percentage every year. If its output doubled in the last two years, find the percentage increase every year.

1.2 Answers – Illustrative Examples (5.1)

Example 1. (i) $2, -\frac{2}{3}$

(ii) $\frac{1}{4}, \frac{1}{4}$

(iii) $\frac{2 \pm 4i}{3}$

Example 2. (i) $\frac{-3 \pm 3\sqrt{3}i}{2}$

(ii) $\frac{5 \pm \sqrt{2}i}{27}$

Example 3. (i) $\frac{\sqrt{2} \pm \sqrt{34}i}{2\sqrt{3}}$
 (ii) $\frac{-1 \pm \sqrt{7}i}{2\sqrt{2}}$

Example 4. $a + 1, -(a + 2)$

Example 5. $2, \frac{1}{2}$

Example 6. $3 + \sqrt{5}i, 3 - \sqrt{5}i$

Example 7. 9, 10, 11

Example 8. 36

Example 9. $100(\sqrt{2} - 1)\%$

1.3 Exercise 5.1

Very short answer / objective questions (1 to 5)

Solve the following quadratic equations:

- (i) $x^2 + 2 = 0$ (ii) $4x^2 + 5 = 0$
- (i) $x^2 - x - 12 = 0$ (ii) $25x^2 + 30x + 7 = 0$
- (i) $\sqrt{3}x^2 - 4x + \sqrt{3} = 0$ (ii) $15x^2 + 15 = 34x$
- (i) $x^2 + x + 1 = 0$ (ii) $x^2 - x + 2 = 0$
- (i) $x^2 + 3x + 5 = 0$ (ii) $2x^2 + x + 1 = 0$

Short answer questions (6 to 8)

- (i) $\sqrt{2}x^2 + x + \sqrt{2} = 0$ (ii) $\sqrt{5}x^2 + x + \sqrt{5} = 0$
- (i) $21x^2 - 28x + 10 = 0$ (ii) $x^2 - 2x + \frac{3}{2} = 0$
- Find the values of x if $p + 7 = 0$, $q - 12 = 0$ and $x^2 + px + q = 0$.

Long answer questions (9 to 17)

- Divide 39 into 2 parts whose product is 338.
- The sum of two numbers is 9 and the sum of their squares is 41. Taking one number as x , form an equation in x and solve it.
- The sum of the squares of two consecutive positive even integers is 340. Find the integers.
- A two digit number is four times the sum and three times the product of its digits. Find the number.
- The hypotenuse of a right angled triangle is 17 cm long and the difference between the other two sides is 7 cm. Find the lengths of the two unknown sides.

14. Originally a rectangle was twice as long as it is wide. When 4 m were added to its length and 3 m subtracted from its width, the resulting rectangle had an area of 600 m^2 . Find the dimensions of the new rectangle.
15. A rectangular garden 10 m by 16 m is to be surrounded by a concrete walk of uniform width. Given that area of the walk is 120 square metres, find its width.
16. Two years ago, a man's age was three times the square of his son's age. In three years time, his age will be four times his son's age. Find their present ages.
17. A piece of cloth costs Rs175. If the piece were 4 m longer and each metre costs Rs5 less, the cost would have remained unchanged. How long is the piece?

1.4 Answers – Exercise 5.1 (Final Answers Only)

1. (i) $\pm\sqrt{2}i$ (ii) $\pm\frac{\sqrt{5}}{2}i$
2. (i) 4, -3 (ii) $-\frac{3+\sqrt{2}}{5}$
3. (i) $\sqrt{3}, \frac{1}{\sqrt{3}}$ (ii) $\frac{3}{5}, \frac{5}{3}$
4. (i) $-\frac{1+\sqrt{5}i}{2}$ (ii) $\frac{1\pm\sqrt{7}i}{2}$
5. (i) $-\frac{3\pm\sqrt{11}i}{2}$ (ii) $-\frac{1+\sqrt{7}i}{4}$
6. (i) $-\frac{1+\sqrt{7}i}{2\sqrt{2}}$ (ii) $-\frac{1\pm\sqrt{19}i}{2\sqrt{5}}$
7. (i) $\frac{14\pm\sqrt{14}i}{21}$ (ii) $\frac{2\pm\sqrt{2}i}{2}$
8. 3, 4
9. 26, 13
10. 4, 5
11. 12, 14
12. 24
13. 8 cm, 15 cm
14. Length = 40 m, Width = 15 m
15. 2 metres
16. 29 years, 5 years
17. 10 metres

2 5.2 Equations Reducible to Quadratic Form

2.1 Illustrative Examples

Example 1. Find the roots of the equation $\frac{1}{x+1} + \frac{2}{x+2} = \frac{4}{x+4}$.

Example 2. Solve: $\frac{x-1}{x-2} - \frac{x-2}{x-3} = \frac{x-5}{x-6} - \frac{x-6}{x-7}$.

Example 3. Solve: $\frac{a}{ax+1} + \frac{b}{bx+1} = a+b$, $ab \neq 0$, $a+b \neq 0$.

Example 4. Solve: $2^{2x+8} + 1 = 32 \cdot 2^x$.

Example 5. Solve: $(x^2 - 5x)^2 - 30(x^2 - 5x) - 216 = 0$.

Example 6. Solve: $(x+1)(x+2)(x+3)(x+4) = 120$.

Example 7. Solve: $x^{2/3} + x^{1/3} = 2$.

Example 8. Solve the equation $8\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = 2$.

Example 9. Solve the equation $\sqrt{a+x} + \sqrt{a-x} = \frac{a}{x}$, $x \neq 0$.

Example 10. Solve $\sqrt{x} = x - 2$.

Example 11. Solve the equation $\sqrt{x^2 + 3x + 32} + \sqrt{x^2 + 3x + 5} = 9$.

Example 12. Solve the equation $\sqrt{x+5} + \sqrt{x+21} = \sqrt{6x+40}$.

Example 13. Solve the equation $\left(x + \frac{1}{x}\right)^2 = 4 + \frac{3}{2}\left(x - \frac{1}{x}\right)$.

Example 14. Solve $4x^4 - 16x^3 + 7x^2 + 16x + 4 = 0$.

Example 15. Solve: $(x+2)(x+3)(x+8)(x+12) = 4x^2$.

2.2 Answers – Illustrative Examples (5.2)

Example 1. $2 + 2\sqrt{3}$, $2 - 2\sqrt{3}$

Example 2. $4\frac{1}{2}$

Example 3. 0 , $-\frac{a^2 + b^2}{ab(a+b)}$

Example 4. $x = -4$

Example 5. 9 , -4 , 3 , 2

Example 6. 1 , -6 , $\frac{-5 \pm i\sqrt{39}}{2}$

Example 7. 1 , -8

Example 8. $x = 1$

Example 9. $a, -a$

Example 10. 4

Example 11. $-4, 1$

Example 12. 4

Example 13. $-\frac{1}{2}, -1, 1, 2$

Example 14. $2, -\frac{1}{2}, \frac{5 \pm \sqrt{41}}{4}$

Example 15. $-4, -6, \frac{-15 \pm \sqrt{129}}{2}$

2.3 Exercise 5.2

Solve the following equations:

1. (i) $\frac{x-2}{x+2} + \frac{x+2}{x-2} = 4$
2. (ii) $\frac{x-1}{x-2} + \frac{x-3}{x-4} = 3\frac{1}{3}$
3. $\frac{1}{x+1} - \frac{2}{x+2} = \frac{3}{x+3} - \frac{4}{x+4}$
4. (i) $3^x + 3^{-x} = 2$ (ii) $2^{x+1} + 4^x = 8$
5. $5^x + 1 + 5^{2-x} = 5^3 + 1$
6. $x^{2/3} - x^{1/3} - 2 = 0$
7. (i) $(x^2 + 3x + 2)^2 - 8(x^2 + 3x) - 4 = 0$
8. (ii) $(x^2 - 5x + 7)^2 - (x - 2)(x - 3) = 1$
9. $x(x+2)(x+3)(x+5) = 72$
10. $x(x+2)(x^2 - 1) = -1$
11. $(x+1)(2x+3)(2x+5)(x+3) = 945$
12. $(x^2 + x - 6)(x^2 - 3x - 4) = 24$
13. (i) $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$
14. (ii) $\sqrt{\frac{4x-1}{4x+1}} - \sqrt{\frac{4x+1}{4x-1}} = \frac{8}{3}, x \neq \pm \frac{1}{4}$
15. (i) $\sqrt{3x+4} = x$ (ii) $\sqrt{3x+1} - \sqrt{x-1} = 2$
16. (i) $\sqrt{x+1} + \sqrt{x-1} = 1$ (ii) $\sqrt{x+3} + \sqrt{x-3} = 6$

17. (i) $\sqrt{3x^2 - 4x + 34} - \sqrt{3x^2 - 4x - 11} = 5$
 18. (ii) $\sqrt{3x^2 - 7x - 30} - \sqrt{2x^2 - 7x - 5} = x - 5$
 19. $12 + 9\sqrt{(x-1)(3x+2)} = 3x^2 - x$
 20. $\frac{\sqrt{x^2+1} - \sqrt{x^2-1}}{\sqrt{x^2+1} + \sqrt{x^2-1}} = \frac{1}{3}$
 21. (i) $\left(x + \frac{1}{x}\right)^2 - 2\left(x - \frac{1}{x} + 4\right) = 11$
 22. (ii) $12x^4 - 56x^3 + 89x^2 - 56x + 12 = 0$
 23. Solve $(x^2 + 2)^2 + 8x^2 = 6x(x^2 + 2)$

2.4 Answers – Exercise 5.2 (Final Answers Only)

1. (i) $2\sqrt{3}, -2\sqrt{3}$ (ii) $5, \frac{5}{2}$
 2. $0, -\frac{5}{2}$
 3. (i) 0 (ii) 1
 4. $-1, 2$
 5. $-1, 8$
 6. (i) $-4, -3, 0, 1$ (ii) $2, 3, \frac{5 \pm \sqrt{3}i}{2}$
 7. $1, -6, \frac{-5 \pm i\sqrt{23}}{2}$
 8. $-\frac{1 + \sqrt{5}}{2}, -\frac{1 - \sqrt{5}}{2}$
 9. $2, -6, \frac{-4 + i\sqrt{59}}{2}$
 10. $0, 1, \frac{1 + \sqrt{57}}{2}$
 11. (i) $\frac{9}{13}, \frac{4}{13}$ (ii) $-\frac{5}{16}$
 12. (i) 4 (ii) $1, 5$
 13. (i) No solution (ii) $\frac{37}{4}$
 14. (i) $3, -\frac{5}{3}$ (ii) $5, 6$
 15. $6, -\frac{17}{3}$

16. $\pm\sqrt{\frac{5}{3}}$
17. (i) $\frac{5 \pm \sqrt{29}}{2}, -\frac{3 \pm \sqrt{13}}{2}$ (ii) $\frac{1}{2}, \frac{2}{3}, \frac{3}{2}, 2$
18. $1 \pm i, 2 \pm \sqrt{2}$

3 5.3 Nature of Roots of a Quadratic Equation

3.1 Illustrative Examples

Example 1. Discuss the nature of the roots of the following equations:

- (i) $4x^2 - 12x + 9 = 0$
 (ii) $3x^2 - 10x + 3 = 0$
 (iii) $9x^2 - 2 = 0$
 (iv) $x^2 + x + 1 = 0$

Example 2. Discuss the nature of the roots of the following equations:

- (i) $4x^2 - 4\sqrt{3}x + 3 = 0$
 (ii) $2\sqrt{3}x^2 - 5x + \sqrt{3} = 0$

Example 3. If $a, b, h \in \mathbb{R}$, show that the equation $(x - a)(x - b) = h^2$ has real roots.

Example 4. If the roots of $ax^2 + x + b = 0$ are real and distinct, show that the roots of the equation $\frac{x^2 + 1}{x} = 4\sqrt{ab}$ are imaginary.

Example 5. Given that $a, b, c \in \mathbb{Q}$, show that the roots of the equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are rational. Find the condition for the roots to be equal.

Example 6. Discuss the nature of the roots of the equation $(m + 6)x^2 + (m + 6)x + 2 = 0$.

Example 7. For what value of k does the expression $x^2 - 2x(1 + 3k) + 7(3 + 2k)$ become a perfect square?

Example 8. Determine a positive real value of k such that both the equations $x^2 + kx + 64 = 0$ and $x^2 - 8x + k = 0$ may have real roots.

Example 9. Show that if p, q, r, s are real numbers and $pr = 2(q + s)$, then at least one of the equations $x^2 + px + q = 0$ and $x^2 + rx + s = 0$ has real roots.

3.2 Answers – Illustrative Examples (5.3)

- Example 1. (i) Rational and equal
 (ii) Rational and unequal
 (iii) Irrational and unequal
 (iv) Pair of complex conjugates

Example 2. (i) Real and equal

(ii) Real and unequal

Example 3. Proven (discriminant ≥ 0)

Example 4. Proven (discriminant of second equation is negative)

Example 5. Roots are rational. Equal when $b = \frac{a+c}{2}$

Example 6.

- Real & equal when $m = 2$ or $m = -6$
- Real & unequal when $m < -6$ or $m > 2$
- Complex when $-6 < m < 2$

Example 7. $k = 2$ or $k = -\frac{10}{9}$

Example 8. $k = 16$

Example 9. Proven

3.3 Exercise 5.3

1. Find the nature of roots of the following equations without solving them:

(i) $x^2 + 9 = 0$

(ii) $4x^2 - 24x + 35 = 0$

(iii) $x^2 - 2\sqrt{2}x + 1 = 0$

(iv) $2x^2 - 2\sqrt{5}x + 3 = 0$

2. (i) Prove that the roots of the equation $x^2 - 2ax + a^2 - b^2 - c^2 = 0$ are always real.

3. (ii) Show that roots of the equation $(x-a)(x-b) = abx^2$ are always real. When are they equal?

4. Show that the roots of the equation $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are always real. Find the condition that the roots may be equal.

5. Discuss the nature of roots of the following equations:

(i) $\sqrt{3}x^2 - 2x - \sqrt{3} = 0$

(ii) $x^2 - (p+1)x + p = 0$

(iii) $(x-a)(x-b) = ab$

6. Find m so that roots of the equation $(4+m)x^2 + (m+1)x + 1 = 0$ may be equal.

7. Show that the roots of the equation $x^2 + 2(3a+5)x + 2(9a^2+25) = 0$ are complex unless $a = \frac{5}{3}$.

8. If $a, b, c, d \in \mathbb{R}$, show that the roots of the equation $(a^2+c^2)x^2 + 2(ab+cd)x + (b^2+d^2) = 0$ cannot be real unless they are equal.

9. If $a, b, c \in \mathbb{Q}$ and $a+b+c = 0$, show that roots of the equation $ax^2 + bx + c = 0$ are rational.

10. Prove that the equation $x^2 + px - 1 = 0$ has real and distinct roots for all real values of p .
11. Show that the roots of the equation $x^2 - 2\left(\frac{m+1}{m}\right)x + 3 = 0$ are real for all (non-zero) real values of m .
12. If the roots of the equation $ax^2 + 2cx + b = 0$ are real and distinct, then show that the roots of the equation $x^2 - 2(a+b)x + a^2 + b^2 + 2c^2 = 0$ are non-real complex numbers.
13. If the roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are equal, show that $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$.
14. For what value of k , $(4-k)x^2 + 2(k+2)x + (8k+1)$ is a perfect square.
15. If the roots of the equation $x^2 - 8x + m(m-6) = 0$ are real and distinct, then find all possible values of m .

3.4 Answers – Exercise 5.3 (Final Answers Only)

1. (i) Pair of complex conjugates
2. (ii) Rational and unequal
3. (iii) Real and unequal
4. (iv) Pair of complex conjugates
5. (ii) Roots are real and equal when $a = b = 0$
6. Roots are equal when $a = b = c$. Then roots are a, a
7. (i) Real and distinct
8. (ii) Rational and distinct when $p \neq 1$; equal when $p = 1$
9. (iii) Real and distinct when $a + b \neq 0$; both zero when $a + b = 0$
10. $m = 5$ or $m = -3$
11. Proven
12. Proven
13. Proven
14. Proven
15. Proven
16. Proven
17. $k = 0$ or $k = 3$
18. $-2 < m < 8$

4 5.4 Relations between Roots and Coefficients

4.1 Illustrative Examples

Example 1. If α, β are roots of the equation $x^2 - 4x + 2 = 0$, find the values of:

- (i) $\alpha^2 + \beta^2$
- (ii) $\alpha^2 - \beta^2$
- (iii) $\alpha^3 + \beta^3$
- (iv) $\frac{1}{\alpha} + \frac{1}{\beta}$

Example 2. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then show that $ax^2 + bx + c = a(x - \alpha)(x - \beta)$.

Example 3. For the quadratic equation $(k - 1)x^2 = kx - 1$, $k \neq 1$, find k so that:

- (i) one root is -3
- (ii) the sum of the roots is 2
- (iii) the product of the roots is -3
- (iv) the roots are equal
- (v) the roots are numerically equal but opposite in sign

Example 4. If one of the roots of $3x^2 - 7px + 10p = 0$ be 5 , find p and also the other root.

Example 5. The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. Prove that the product of the roots is $-\frac{1}{2}(a^2 + b^2)$.

Example 6. If α, β be the roots of the equation $ax^2 + bx + c = 0$, evaluate:

- (i) $\alpha^3\beta + \alpha\beta^3$
- (ii) $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$
- (iii) $(a\alpha + b)^2 + (a\beta + b)^2$

4.2 Answers – Illustrative Examples (5.4)

Example 1. (i) 12
 (ii) $\pm 8\sqrt{2}$
 (iii) 40
 (iv) 2

Example 2. Proven

Example 3. (i) $k = \frac{2}{3}$
 (ii) $k = 2$
 (iii) $k = \frac{2}{3}$

(iv) $k = 2$

(v) $k = 0$

Example 4. $p = 3$, other root = 2

Example 5. Proven

Example 6. (i) $\frac{c(b^2 - 2ac)}{a^3}$

(ii) $\frac{b(b^2 - 2ac)}{a^2c}$

(iii) $\frac{b^2(b^2 - 2ac)}{a^2}$